

CHAPTER: 3

Oligopoly Market

Chapter outline

- Introduction
- Characteristics
- Causes
- Equilibrium
 - Non collusive oligopoly firms
 - Cournot's duopoly model
 - Kinked demand model
 - Bertrand model
 - Stackelberg's leadership model
 - Collusive oligopoly models
 - Cartel
 - Price leadership

Introduction

- a. Definition:** A market structure in which a few firms dominate the supply of an industry's output and compete with each other for markets
- The main difference b/n oligopoly and other markets is the existence of *strategic interdependence* of firms
 - In perfect competition and pure monopoly price and output decision of one firm do not affect other firms decision
 - In oligopoly it affects decision of the other firm

b. Characteristics

- ***Few number*** of firms in a given industry
 - the limiting case of oligopoly market is where there are only two firms and this is called *duopoly*
- Firms produce ***homogenous or differentiated*** product
- ***Interdependence*** of firms in decision making
- Firms ***have some power*** to set price
- Firms have ***profit maximizing*** objective

Example (domestic firms like Beer industry, sealed water industry,...)

c. Causes of Oligopoly

- **Economies of scale**
 - only few firms reach minimum of LAC
 - firms with minimum LAC hinder entrances
- **Barriers to entry**
 - Only strong firms successfully pass all the barriers inter the market
- **Collusion (merger of small firms)**
 - Merging of firms create few powerful firms and others removed from the market.
- **These causes reduce number of firm in the market**

d. Equilibrium of a firm

- There is no standard equilibrium solution in this market
- We study several models of equilibrium
- Two basic categories
 - Collusive models and non collusive models
 - Collusive= cooperation or agreement
 - Non-collusive =no cooperation(coordiantion)

Classification of firms

- Based on the reaction pattern, firms can be classified into

A. Non collusive models

- Each firm makes decision **independently** by making some assumption about its competitors decision.
- No contractual agreement and cooperation in making optimal decision.
- Non-collusive oligopoly models include
 - I. **Kinked demand model**
 - II. **Cournot's duopoly model**
 - III. **Bertrand's duopoly model**
 - IV. **Stackelberg's duopoly model**

B. Collusive Oligopoly

- Firms have **implicit** or **explicit** agreement in making optimal decisions.
- **Why collusion?**
 - collusion increases profit.
 - It also decreases uncertainties.
 - it can act as a barrier to entry of new firms.

Collusive models

- There are two groups of models of collusive oligopoly.

A. Cartel

- i. Cartels aiming at joint profit maximization.
- ii. Market sharing cartels.

B. Price leadership

- i. Price leadership by the low cost firm.
- ii. Price leadership by the dominant firm.
- iii. Barometric price leadership.

A. Non- collusive

i. The kinked demand curve model

- **Developed by Paul Sweezy in 1939.**
- The model advocates that the ***behavior of oligopolistic organizations remain stable*** when the price and output are determined.
- Price generally adjusts to change in demand and supply.
- Example if , for a normal goods, if income of the consumer increase → dd shift out → Price increases.
- If cost of production decreases supply curve shifts right ward and price decreases.
- **But in in some oligopoly model price is sticky (stable)**
due to interdependency of decision and absence of agreement.
- This implies that an oligopolistic market is characterized by a ***certain degree of price rigidity or stability***, especially when there is a change in prices in downward direction.

Kinked Demand modelCont'd

- The model assumes that once equilibrium P is set, firms are reluctant (unwilling) to change P
- If it increases price slightly, others will not follow
 - a slight increase in P leads to high loss of market share ($\% \Delta P < \% \Delta Q$).
 - **so planned demand (dd) is the elastic one**
 - Therefore, the firm is unwilling to increase P above equilibrium P

A. Non- collusive Cont'd

- If it decreases P others also do the same and leads to price war that decreases profit of all firms,
 - by decreasing P it actually sells less than the planned
 - a decrease in P increases sale by small amount ($\% \Delta P > \% \Delta Q$) so,
 - **thus its demand is the less elastic one (DD)**
 - since decrease in price decreases profit of all firms a firm is unwilling to decrease P

A. Non-collusive cont'd

- So the demand curve of a firm is **kinked** at equilibrium P , for above equilibrium P **planned demand curve** and for below equilibrium P the **actual demand curve**.
- MR curve is **discontinuous** at the kink point.
- MC curve passes through the discontinuous gap
- Equilibrium is the kinked point even if MR is not cutting MC curve.
- **Price remains the same as far as the MC varies within the discontinuous range (MC1-MC3)**

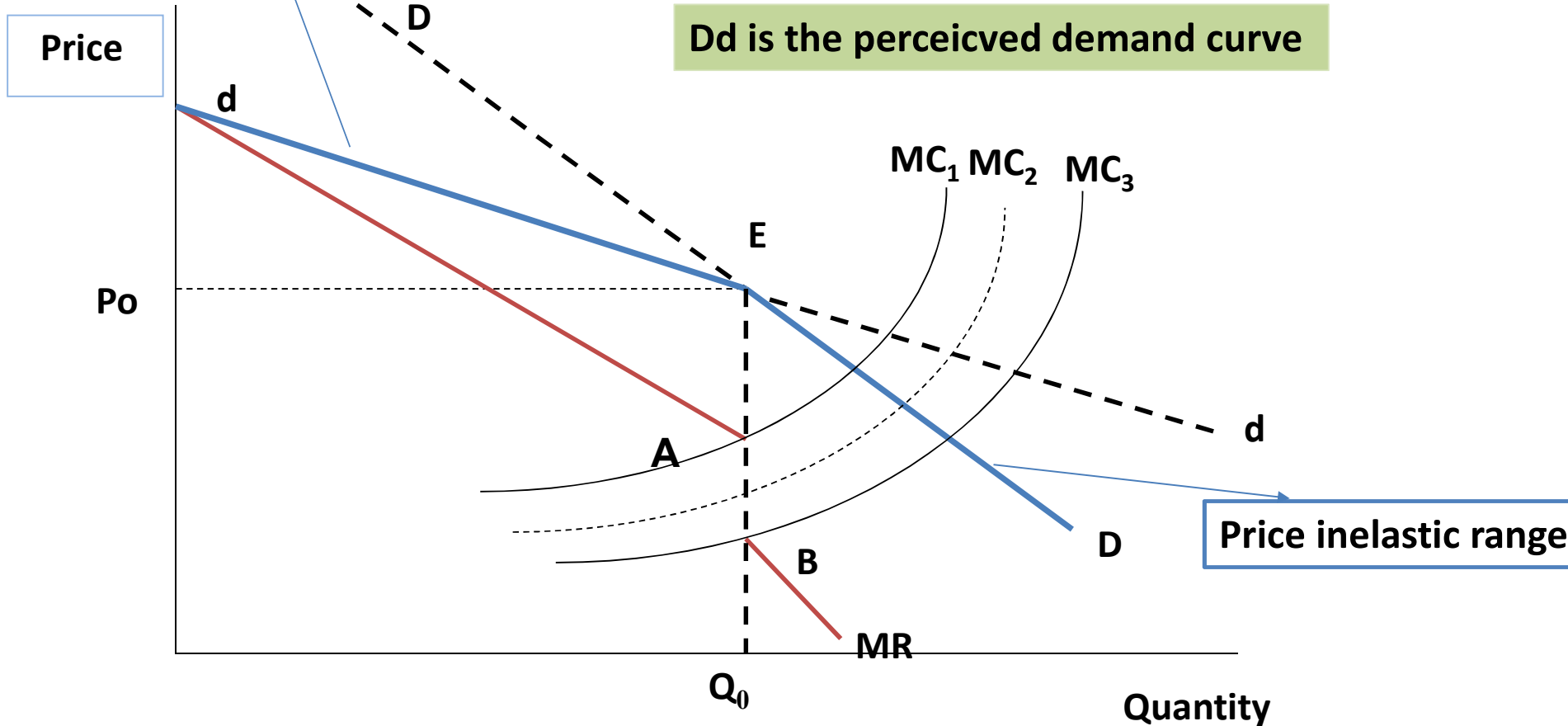
A. Non-collusive cont'd

Price elastic range

The demand curve is dED

DD market share demand curve

Dd is the perceived demand curve



Critics: The model does not show equilibrium P is set

Figure 3.1: The kinked demand curve

ii. Cournot Duopoly Model

- **History**

- Developed by **Antoine Augustin Cournot** (1801–1877).
- companies compete **on the amount of OUTPUT they will produce not based on PRICE.**

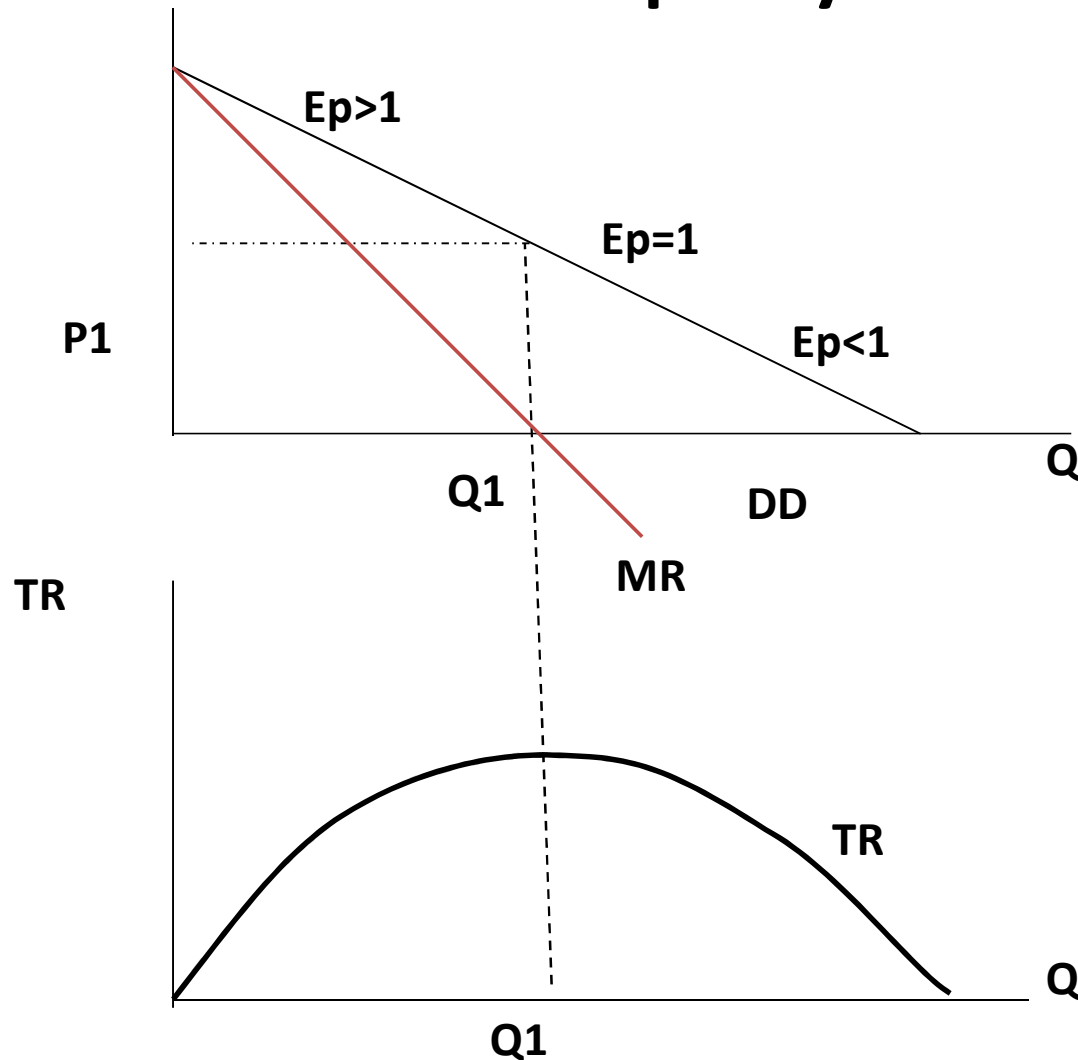
Cournot model

...cont'd

Assumptions

- Two firms (A and B)
- Homogeneous product (mineral water)
- Equal (identical and constant) costs
 - (zero marginal cost)
- Naïve assumptions (firms do not learn from their past experience)
- Linear market demand curve
- Equilibrium price is set by the market
- Firms know that output of one firm affects output of the other through affecting equilibrium P

ii. Cournot Duopoly Model cont'd ...



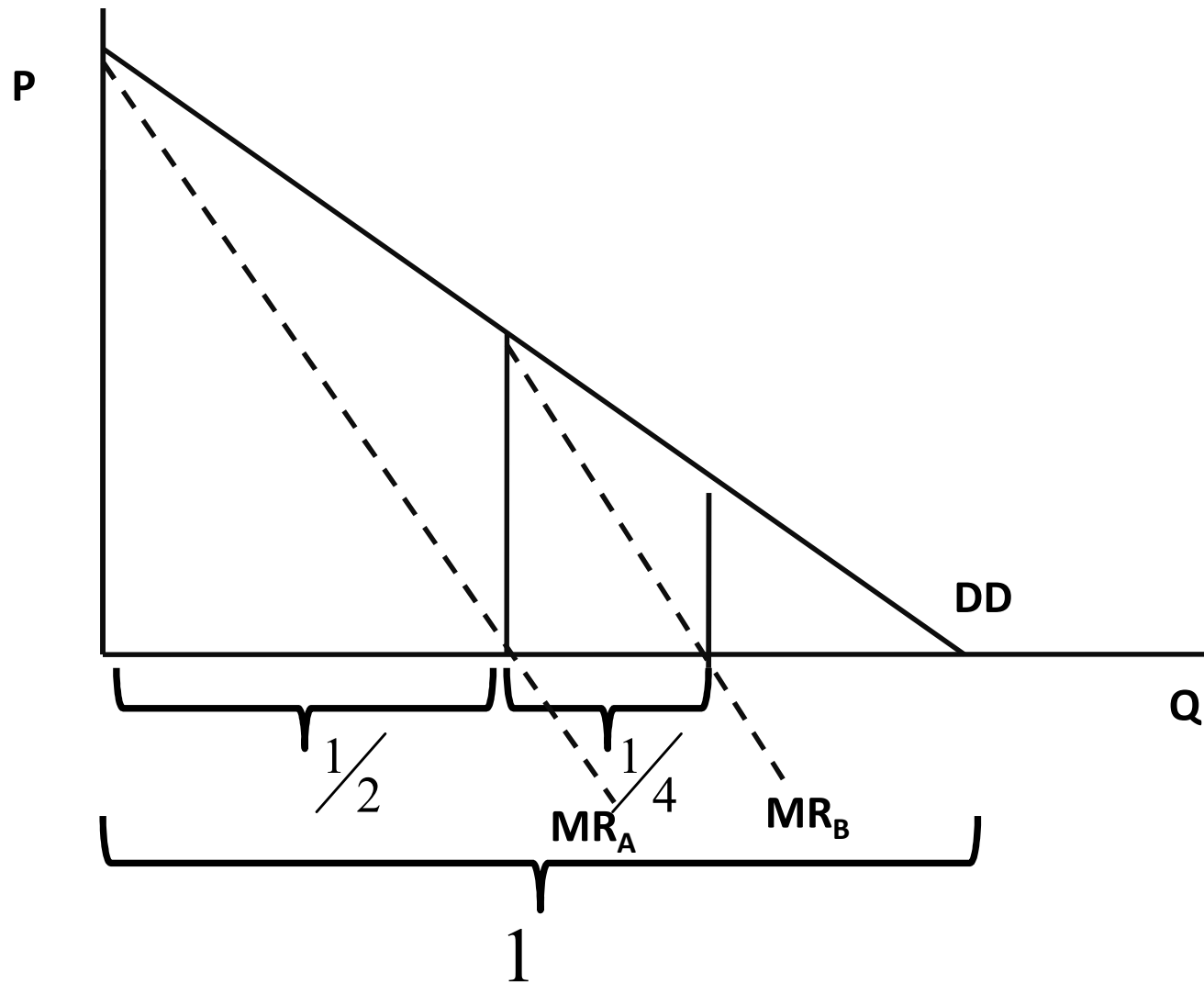
- At equilibrium, **$MR=MC$** , but **$MC=0$** thus where **$MR=0$** ,
- We are at the midpoint (half) of market demand.

Equilibrium quantity of each firm

Period 1:

- **Assume firm A is the first to enter the market**
 - Effective demand curve is the market demand
 - Maximize its profit at $MR=MC$, but $MC=0$, thus where $MR=0$
 - $MR=0$, at mid-point (half) of market demand
- **Assume Firm B enter the market**
 - It take unsupplied part of market demand as its demand curve and set its profit maximizing output will be where
 - **$MR=0$** , at mid-point of its DD,
 - it is $QB = 1 \frac{1}{2} \left(1 - \frac{1}{1}\right) = \frac{1}{4}$ of market DD.

Continued...



Continued...

Period 2:

- Now firm A become aware of the existence of other firm, and as it is **producing** $\frac{1}{4}$ of the market DD.
- So its DD is not the market DD but $\frac{3}{4}$ of DD
- Maximize by producing half of what is not supplied by B.
 - $QA = \frac{1}{2} \left(1 - \frac{1}{4} \right) = \frac{3}{8}$ of market DD.
- Firm B become aware that A is producing $\frac{3}{8}$ of market DD
- Thus it maximize by supplying $\frac{1}{2}$ of what is not supplied by A.
 - $QB = \frac{1}{2} \left(1 - \frac{3}{8} \right) = \frac{5}{16}$ of market DD.

Continued...

Period 3:

- Firm A adjust again by producing $\frac{1}{2}$ of what is not supplied by B
- Firm B too, adjust again by producing $\frac{1}{2}$ of what is not supplied by A
- The process continued in the same way until each firm produce $\frac{1}{3}$ market DD at equilibrium

Continued....

Period 1

- $A = \frac{1}{2}$ $B = \frac{1}{2} (1 - \frac{1}{2}) = \frac{1}{4}$

Period 2

- $A = \frac{1}{2} (1 - \frac{1}{4}) = \frac{3}{8}$ $B = \frac{1}{2} (1 - \frac{3}{8}) = \frac{5}{16}$

Period 3

- $A = \frac{1}{2} (1 - \frac{5}{16}) = \frac{11}{32}$... $B = \frac{1}{2} (1 - \frac{11}{32}) = \frac{21}{64}$

Period 4

- $A = \frac{1}{2} (1 - \frac{21}{64}) = \frac{43}{128}$.. $B = \frac{1}{2} (1 - \frac{43}{128}) = \frac{85}{256}$
- Period continues

Equilibrium output of A

$$A = \frac{1}{2}, \frac{3}{8}, \frac{11}{32}, \frac{43}{128} \dots \dots \dots = \frac{1}{3}$$

$$A = \frac{1}{2} - \frac{1}{8} - \frac{1}{32} - \frac{1}{128} \dots \frac{1}{4}$$

$$A = \frac{1}{2} - \frac{1}{8} - \frac{1}{8} \frac{1}{4} - \frac{1}{8} \frac{1}{4^2} - \frac{1}{8} \frac{1}{4^3} \dots$$

$$A = \frac{1}{2} - \left(\frac{1}{8} + \frac{1}{8} \frac{1}{4} + \frac{1}{8} \frac{1}{4^2} + \frac{1}{8} \frac{1}{4^3} \dots \right) \frac{1}{8}$$

- Geometric progression formula

$$G = \frac{G_1}{1 - r} \quad \text{Where -G1 is first term of the progression-}$$

- r is the common ratio -

$$A = \frac{1}{2} - \left(\frac{\frac{1}{8}}{1 - \frac{1}{4}} \right) = \frac{1}{3}$$

Equilibrium Out put of B

$$B = \frac{1}{4}, \frac{5}{16}, \frac{21}{64}, \frac{85}{256}, \dots$$

$$B = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \frac{1}{256} + \dots$$

$$B = \frac{1}{4} + \frac{1}{4} \frac{1}{4} + \frac{1}{4} \frac{1}{4}^2 + \frac{1}{4} \frac{1}{4}^3 + \dots$$

$$G = \frac{G}{1 - r}, B = \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{1}{3}$$

General formulan number of firms

- A market share of n number of firms in the market is measured by $q_n^* = \frac{1}{n+1}$.
- Example 1; if total demand of market at zero price is **120** units and **5** firms are there in the market, what is market share of each firm and how much does a firm supply?
- *Share* = $q_n = \frac{1}{n+1} = \frac{1}{5+1} * 120 = 20 \text{ units}$
- Supply of each = $\frac{1}{n+1} = \frac{1}{6}$.

Continued....

- Example 2
 - Assume there are **four** firms in the market and the market demand equation is given by **$P = 8000 - 2Q$**
- Then;
 - a) find market share of the **two** firms.
 - b) how much each firm supply (hint: $p=0$)
- Answer
 - *0.4 or 2/5.*
 - **$q_n = 800.$**

Cournot's Reaction Curve Approach

Assumptions

- Two firms (A&B)
- Homogeneous Product
- **Different cost**
 - The assumption of costless production is unrealistic.
- Both firms are **naïve** and set equilibrium quantity **simultaneously** assuming output of the other is constant
- **Equilibrium Price is set by the market**
- Price set by amount of output of A and B ($Q=Q_A+Q_B=SS$)
i.e. Change in either of Q_A or Q_B or both affect SS and change equilibrium P because P is set where $DD=SS$

Continued ...

- **Each firm, assuming maximum output of the other is given, determine its output that maximize its profit**

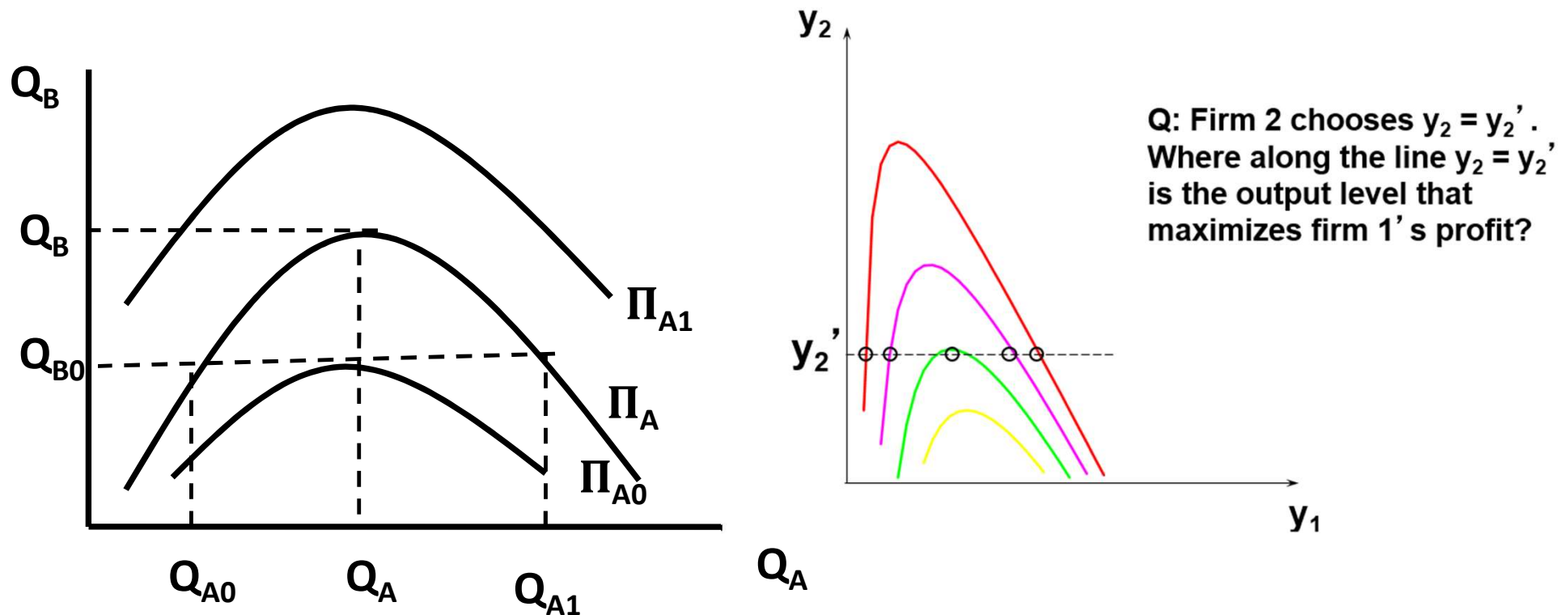
$$\Pi_A = f(PQ_A)$$

$$\Pi_B = f(PQ_B)$$

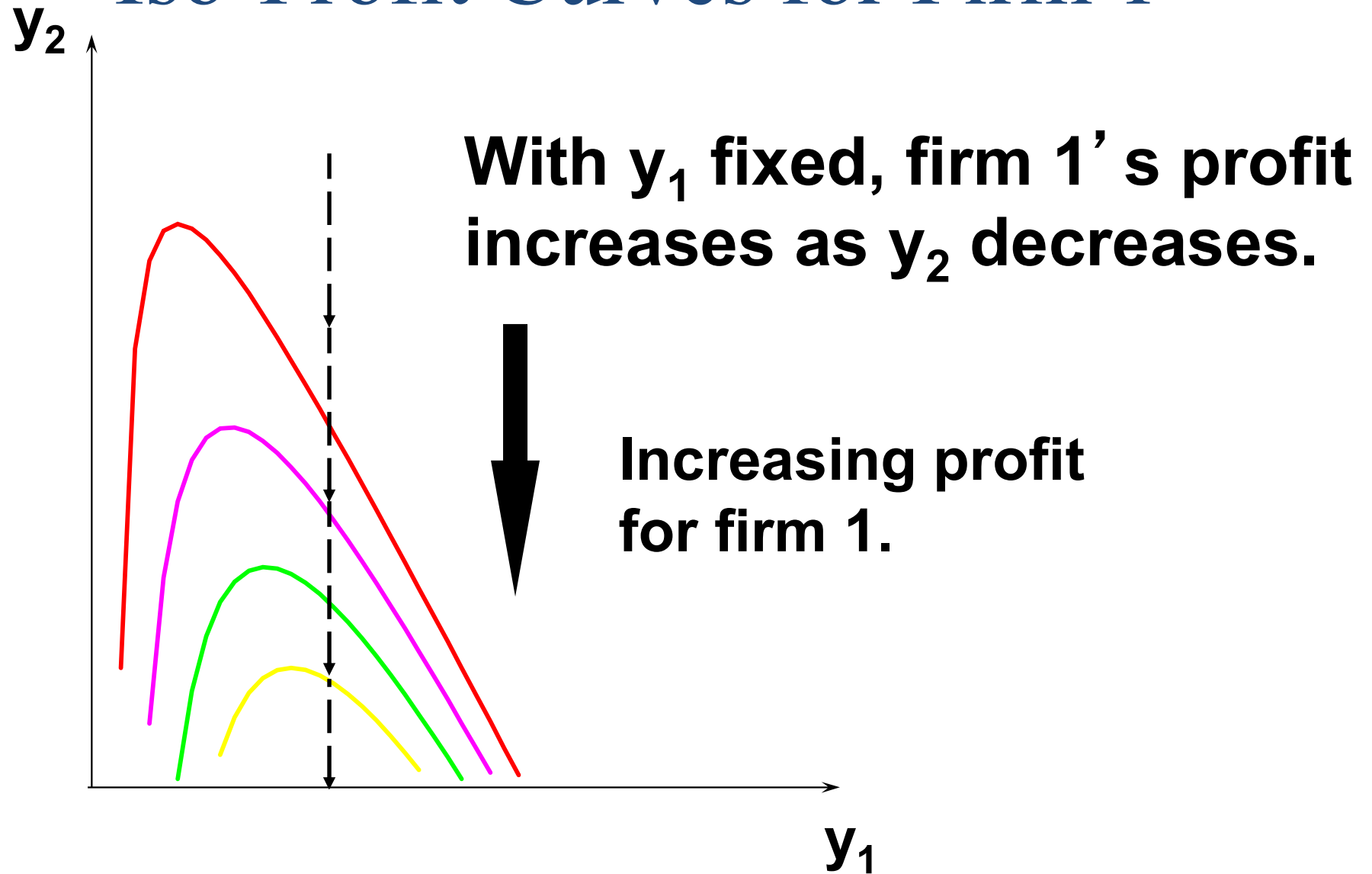
- Example Firm A sets Q_A by assuming Q_B is maximum
- If firm A expect that firm B will decrease Q_B to Q_{B0} , that affect profit of both firms, firm A can maintain its profit by constant by increasing Q_A or decreasing Q_A
- First firm A get Π_A from Q_A combined with Q_B
- If B decrease $Q_B \rightarrow$ SS decrease $\rightarrow$ P increase.
- So firm A can get the **same profit** (Π_A)
 - i) By producing less (Q_{A0}) at higher P or
 - ii) By producing more (Q_{A1}) that decrease P

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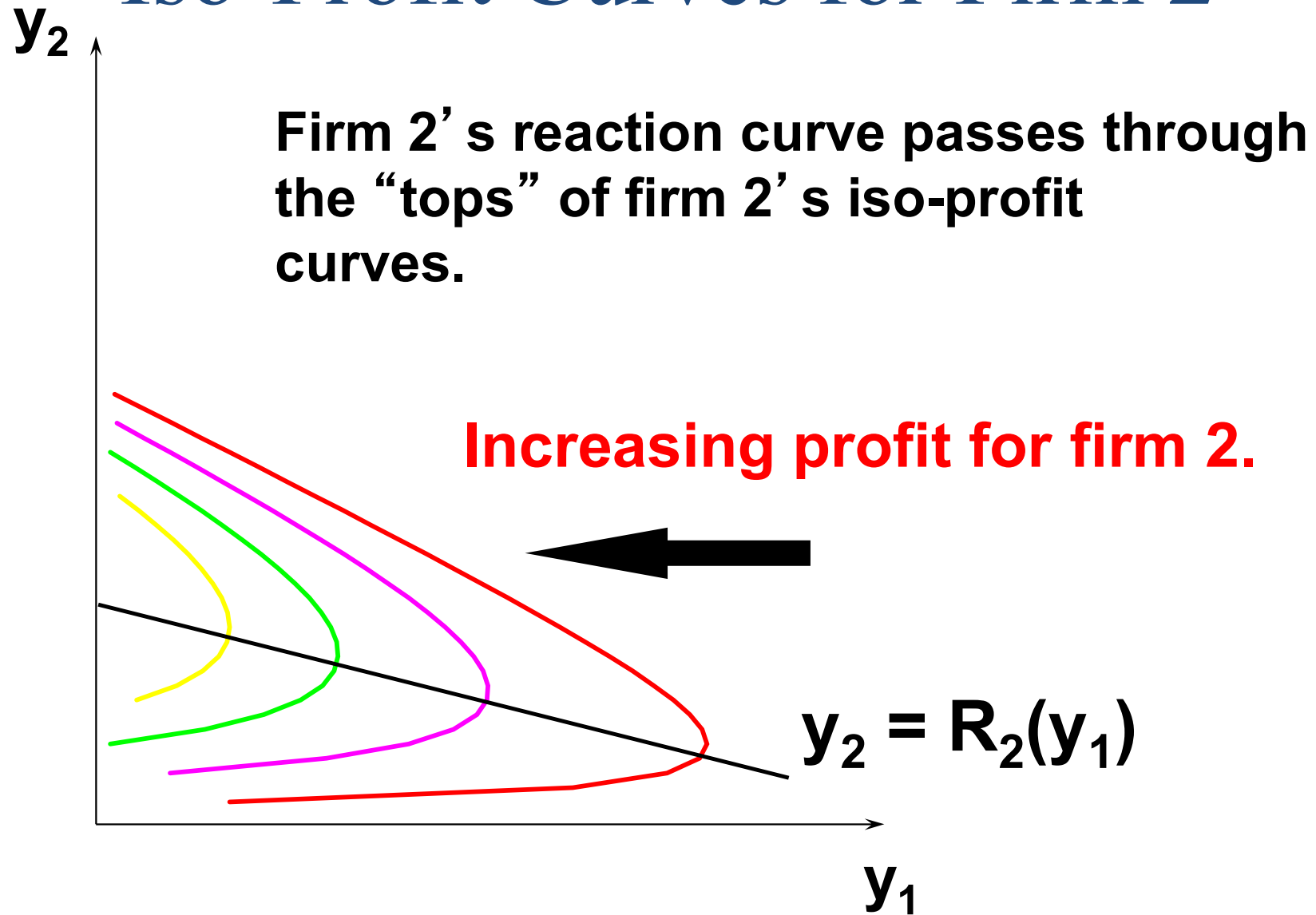
- The curve along which the same amount of profit is derived from different combination of quantity of both firms is called **Isoprofit** curve



Iso-Profit Curves for Firm 1



Iso-Profit Curves for Firm 2



Continued ...

- On the other hand, firm A can react in other way that increase or decrease its profit

Example

- If maximum Q_B that A expect decrease to Q_{B0} and Q_A is constant, firm A get higher profit on lower isoprofit (Π_{A1}) $\uparrow \rightarrow$ because $\Pi_A = P Q_A$
- If maximum Q_B that A expect increase to Q_{B1} and Q_A is constant, firm A get lower profit on highr isoprofit (Π_{A1}) $\downarrow \rightarrow$ Because $\Pi_A = P Q_A$

Continued....

- Optimal point of A is where the expected maximum quantity of B is combined with quantity of A on Isoprofit curve.
- I.e. at the pick point of isoprofit curve.
- That is because firm A naïvely assumed firm B produce Q_B , and it believe that B cannot produce less than or more than the expected
- The optimal point show how firm A react to the maximum out put of B

Properties of Isoprofit

- 1) Isoprofit of a firm for substitute goods is concave to the axis along which we measure output of the firm.
- 2) Isoprofit that lie far from the axis along which we measure output of the firm represent **less profit**.

3) Assumption

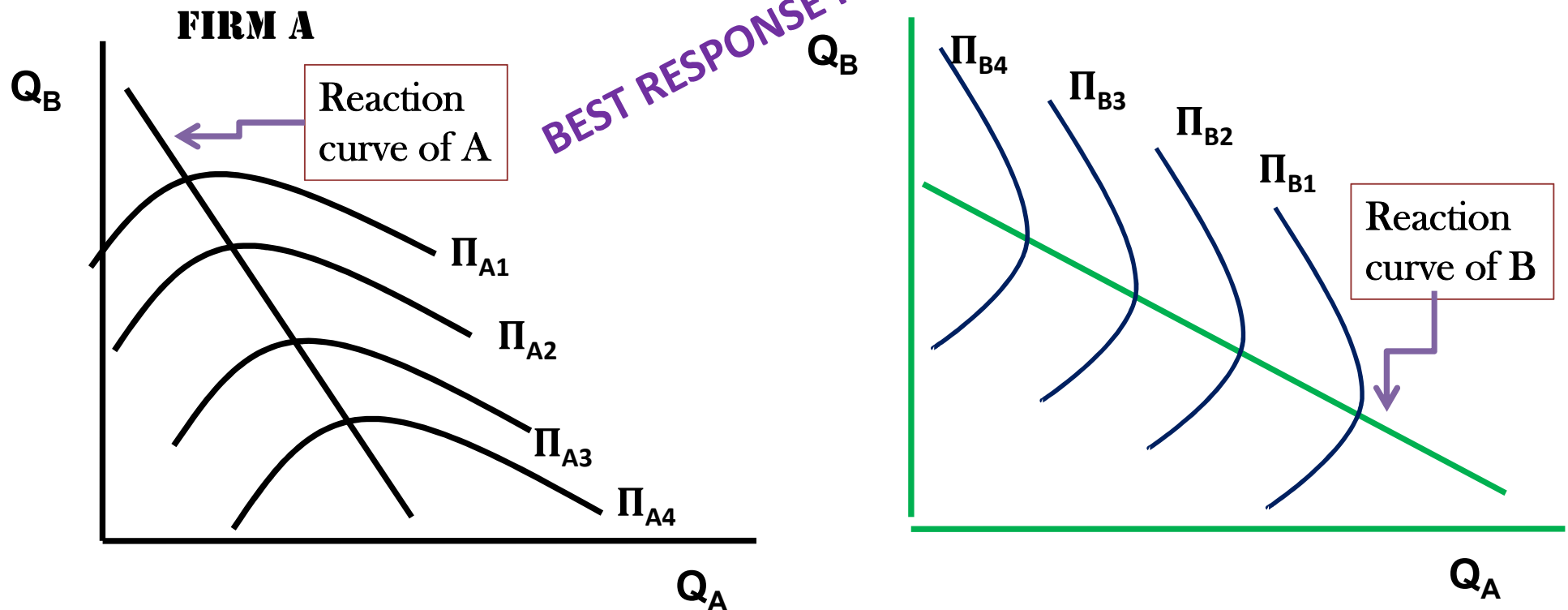
- The pick point (maximum point) of successive isoprofit for one firm lies to the right of each other and the other is to the left of each other.

Example; for A to the left & for B to the right

- By connecting the successive highest point of each isoprofit **reaction curve** is derived

Continued...

Reaction curve: is a locus of highest point of Isoprofit that tell us how much to produce given the output of its competitor.



Equilibrium of both firms

- **Equilibrium is where the reaction curve of both firms intersect.**
- At the intersection point actual production of a firm and the output estimated (assumed) by the other firm are equal.
- At equilibrium profit of each firm maximized, but industry's joint profit is not maximized because of naïve assumption (they don't learn from past experience)
- Had they learn from past experience they make agreement and can get the same profit by producing less output on the contract curve

Equilibrium of the model

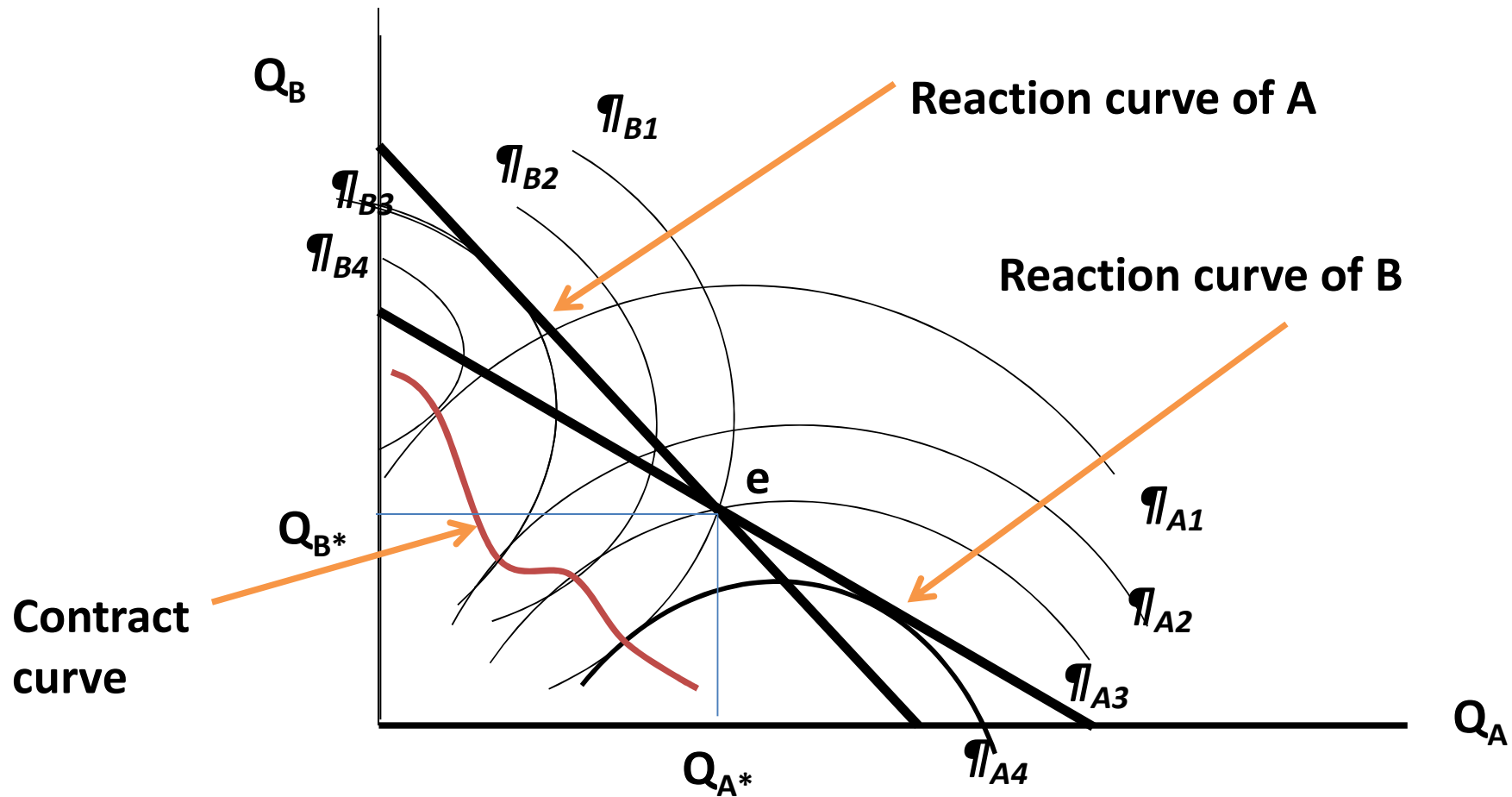
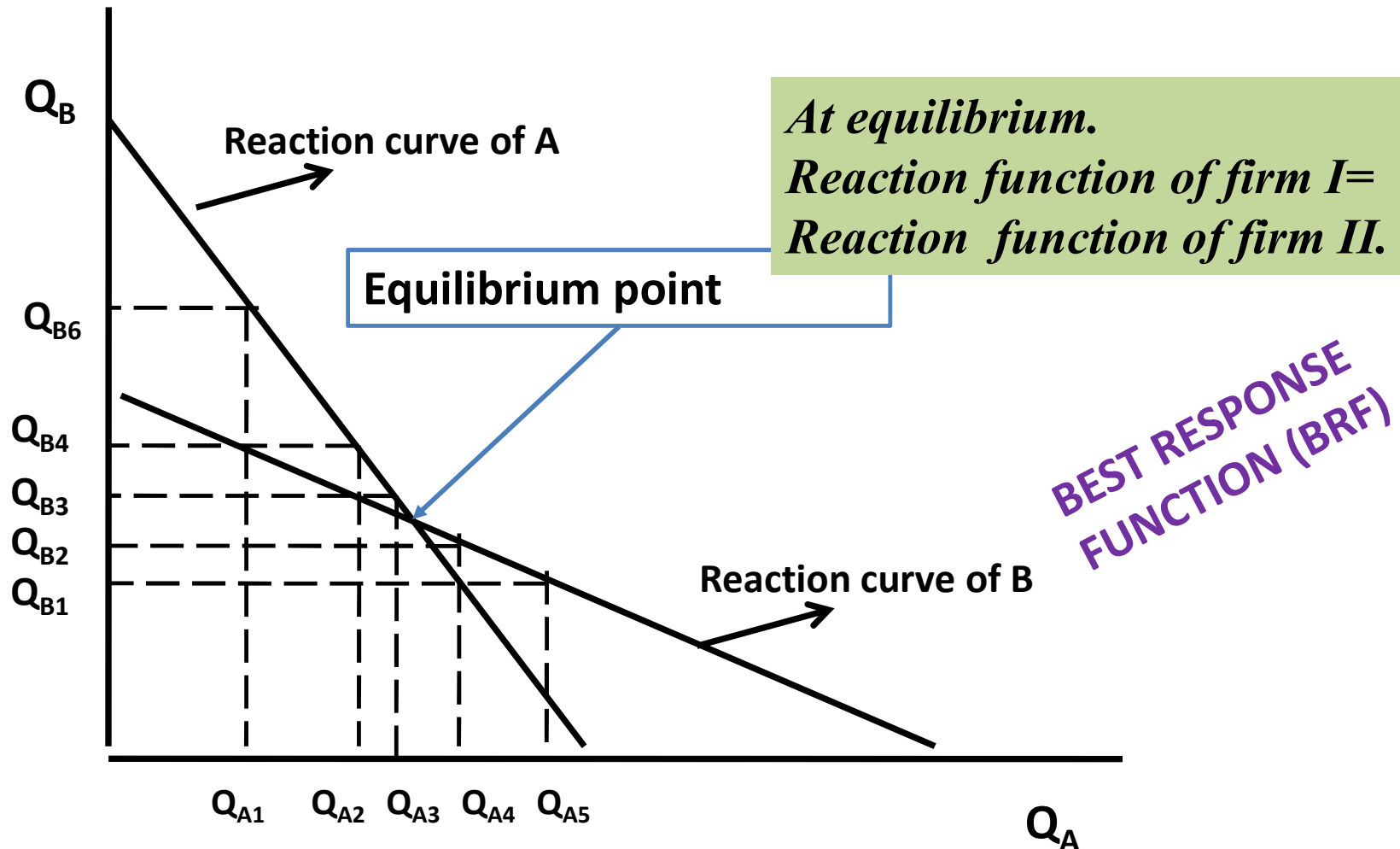


Figure 2.4.: cournot's duopoly model equilibrium

Reaction of firms



Continued....

- Suppose firm A assumed out put of B is Q_{B6} and produced Q_{A1} on its reaction curve (Q_{A1}, Q_{B6})
- But (Q_{A1}, Q_{B6}) is not on reaction curve of B so firm B produce Q_{B5} to make combination (Q_{A1}, Q_{B5}) on its reaction curve.
- Again (Q_{A1}, Q_{B5}) is not on reaction curve of A and A react by increasing production to Q_{A2} on its reaction curve
- The process continues until the combination of the two firms out put become at the point of intersection

Mathematically

- $\Pi_A = PQA - CA \dots$ Is Isoprofit curve equation of A
- $\Pi_B = PQB - CB \dots$ Is Isoprofit curve equation of B

- $\frac{\partial \pi_A}{\partial Q_A} = 0$ gives you reaction curve equation of A.

- $\frac{\partial \pi_B}{\partial Q_B} = 0$ gives you, reaction curve equation of B

❖ At Equilibrium

$$\frac{\partial \pi_A}{\partial Q_A} = \frac{\partial \pi_B}{\partial Q_B}$$

Traditional Example:Cournot

- Two firms selling spring water
- *Inverse demand*

$$- P(Q) = a - Q$$

- Cost

$$- TC_i(q_i) = cq_i$$

– **Equilibrium Outcomes**

$$\bullet q_1^* = q_2^* = \frac{a-c}{3}$$

$$\bullet P^* = \frac{a+2c}{3}$$

$$\bullet \pi^* = \frac{2(a-c)^2}{9}$$

Numerical Example 1: Cournot's model

- Suppose there are two firms supplying homogeneous product
- Market demand $P(Y)=60-2Y$
- **Cost of firm 1 ... $C1=y_1^2$**
- **Cost of firm 2... $C2= 4y_2$** , where Y is output (Q) and $Y=y_1+y_2$ and P is function of Y (price depends on out put)
- Then....

Continued...

- Required:
 - a) Find the isoprofit function of firm 1 and firm 2.
 - b) Find the reaction curve function of firm 1 and firm 2.
 - c) Equilibrium y_1 and y_2
 - d) Equilibrium P
 - e) Show graphically
 - f) The equilibrium profit of both firms and industry

Numerical Example 2: Cournot's model

- Suppose that the market inverse demand function is
- $P(y) = 60 - Q$
- and that the firms total cost functions are
 $c_1(y_1) = y_1^2$ and $c_2(y_2) = 15y_2 + y_2^2$.

Continued...

- Required:
 - a) Find the isoprofit function of firm 1 and firm 2.
 - b) Find the reaction curve function of firm 1 and firm 2.
 - c) Equilibrium y_1 and y_2
 - d) Equilibrium P
 - e) Show graphically
 - f) The equilibrium profit of both firms and industry

Continued....

Solution

a) $\Pi_1 = 60Y_1 - 3Y_1^2 - 2Y_1Y_2$

$$\Pi_2 = 56y_2 - 2y_1y_2 - 2y_2^2$$

b) $y_1 = 10 - \frac{1}{3}y_2 \dots \dots \dots (1)$

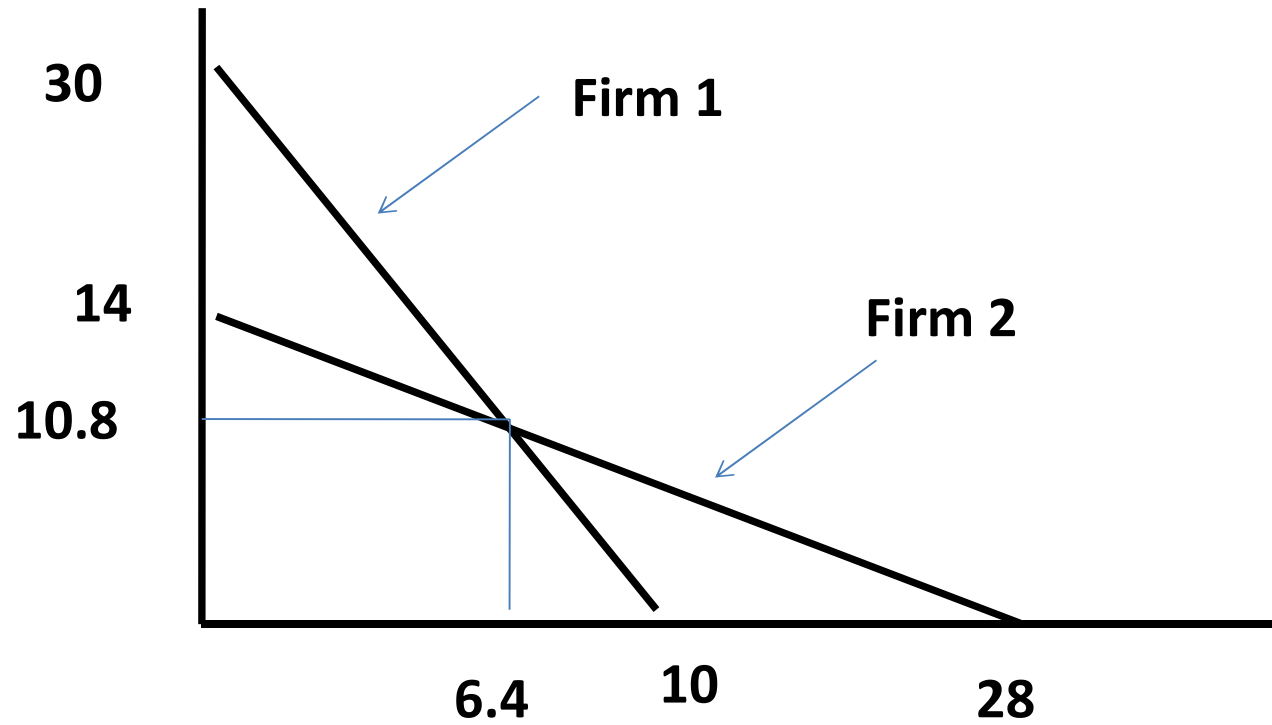
$$y_2 = 14 - 0.5y_1 \dots \dots (2)$$

c) **y1=6.4, y2=10.8, Y=17.2**

Continued...

d) $P=25.6$

e)



f) $\Pi_1 = 163.84$

$\Pi_2 = 233.28$

$\Pi = 397.12$

The Bertrand Model

- The Cournot model takes the firm's strategy space as being quantities.
 - The market determines the price.
- *The Bertrand model considers price as the relevant strategic variable.*
 - *The market determine the quantity sold.*
- Joseph Bertrand presented his model in a review of Cournot's work.
- Joseph Louis François Bertrand (1822 – 1900) was a French mathematician.
- When a firm chooses its price, it must forecast the price set by the other firm in the industry.
- *It involves two identical firms choosing prices simultaneously for their identical products.*
- *Independent and simultaneous decision making*

Continued...

Assumptions.

- **Two firms (A and B).**
- **Homogeneous product.**
- **Identical cost.**
- **Price competition**
- **Due to the homogeneity of the product, each firm face the same demand curve.**
- **Constant Marginal Cost (MC):** Both firms have the same constant marginal cost of production, Firms are naïve and set price simultaneously assuming Price of the other is constant.

The Bertrand Model

- Suppose that we have two firms with constant marginal costs of c that face a market demand curve of $D(p)$.
- Due to the homogeneity of the product, consumers prefer product of the firm **with lower price**.
- As before, we assume a homogeneous product so that the demand curve facing firm 1, say, is given by:

$$d_1(p_1, p_2) = \begin{cases} D(p_1) & \text{if } p_1 < p_2 \\ D(p_1)/2 & \text{if } p_1 = p_2 \\ 0 & \text{if } p_1 > p_2 \end{cases}$$

- That is, firm 1 believes that it can capture the entire market by setting a price smaller than the price of firm 2.
 - Of course, firm 2 is assumed to have similar beliefs.
- ***Undercutting price of your rival firm***

Equilibrium Outcome

- **Scenarios**

- If firm A charges less than firm B, it supply the entire market and B supply nothing
- If firm A charges equal price with B, both supply half of the market
- If firm A charges higher price than firm B, firm A supply nothing and B supply the entire market.

- *if $P_1 < P_2 \Rightarrow Q_1 = Q$*
- *if $P_1 > P_2 \Rightarrow Q_1 = 0 \text{ \& } Q_2 = Q$*
- *if $P_1 = P_2 \Rightarrow Q_1 = Q_2 = \frac{1}{2}Q$*

– Profit of A

$$\pi(P_1, p_2) =$$

$$\left\{ \begin{array}{l} P_1 Y - C(Y) = \pi \\ [P_1 Y - C(Y)] \frac{1}{2} = \pi \frac{1}{2} \\ P_1 0 - C(0) = 0 \end{array} \right.$$

Bertrand Equilibrium

- The problem
 - we want to find a pair of prices such that each price is a profit-maximizing choice given the choice made by the other firm.
- A Bertrand competitor, it maximizes
 - $TR = \{a_1 - b_1 p_1 + c p_2\}(p_1)$, assuming $TC=0$
- For Cournot competitor, the maximization is
 - $TR = \{\alpha_1 - \beta_1 y_1 - \gamma y_2\}(y_1)$, assuming $TC=0$

Equilibrium Outcome

- **Firms try to charge less than the other firm by decreasing price.**
- But firms do not charge less than MC which leads to negative profit.
- **Assume Price of both firms are above MC**
 - If firm A lowers its product price all consumers slowly shift to A and B sell nothing.
 - Then B reacts by lowering its product price to less than price of product of A.
- **The process (action and reaction) continues until price of both firms equal to MC.**

- ✓ **For similar cost @Equilibrium= $P=MC$**
- **The outcome is similar to competitive outcome**
- **Zero profit for each firm**
- **Bertrand Paradox:** Even with only two firms, price competition leads to the same outcome as perfect competition.

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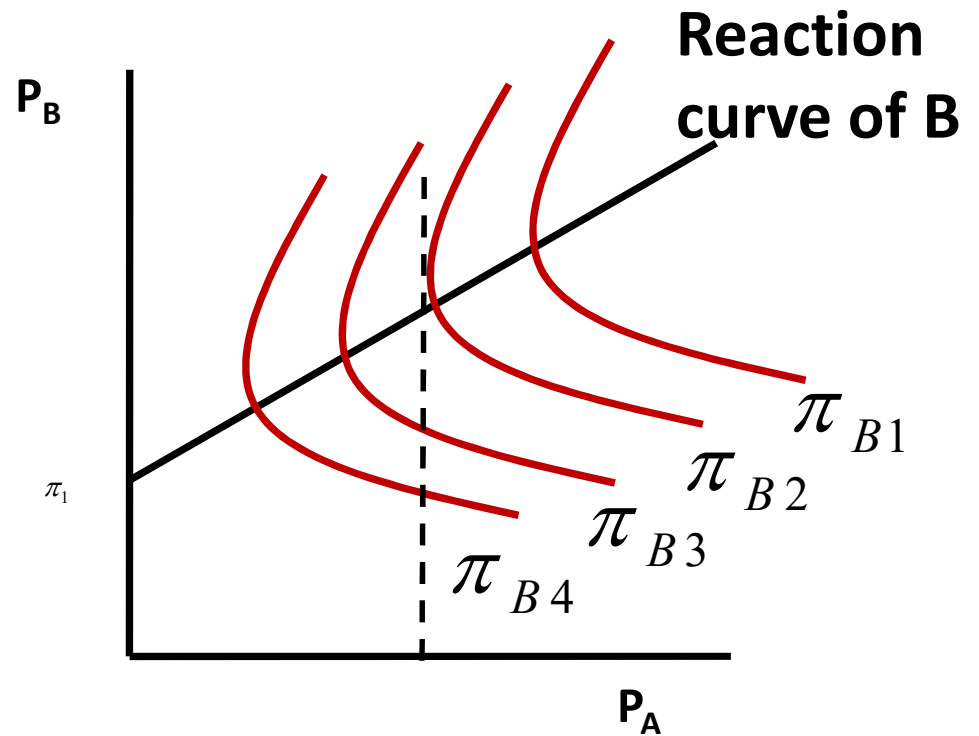
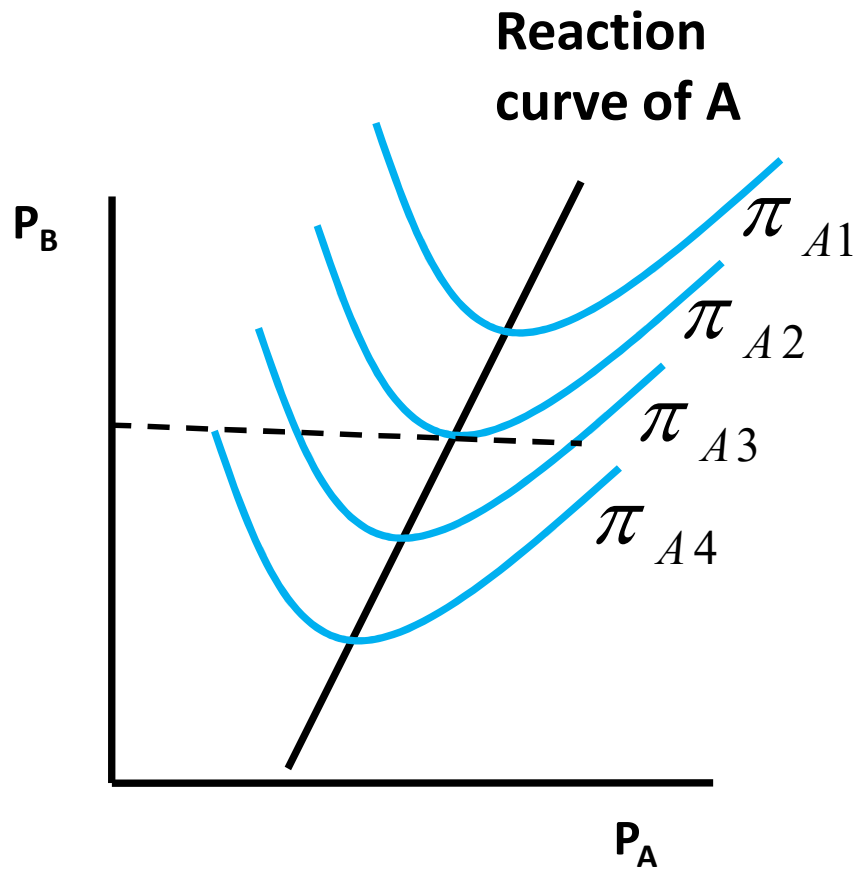
- price greater than marginal cost cannot be stable equilibrium because each firm has an incentive to cut price as long as production remain profitable.
- **The equilibrium can be shown using reaction curve that are derived from Isoprofit curves.**
- ***Isoprofit is a curve that show different combination of price of the two firms that yield equal profit to a firm.***

Continued....

Properties of isoprofit curve

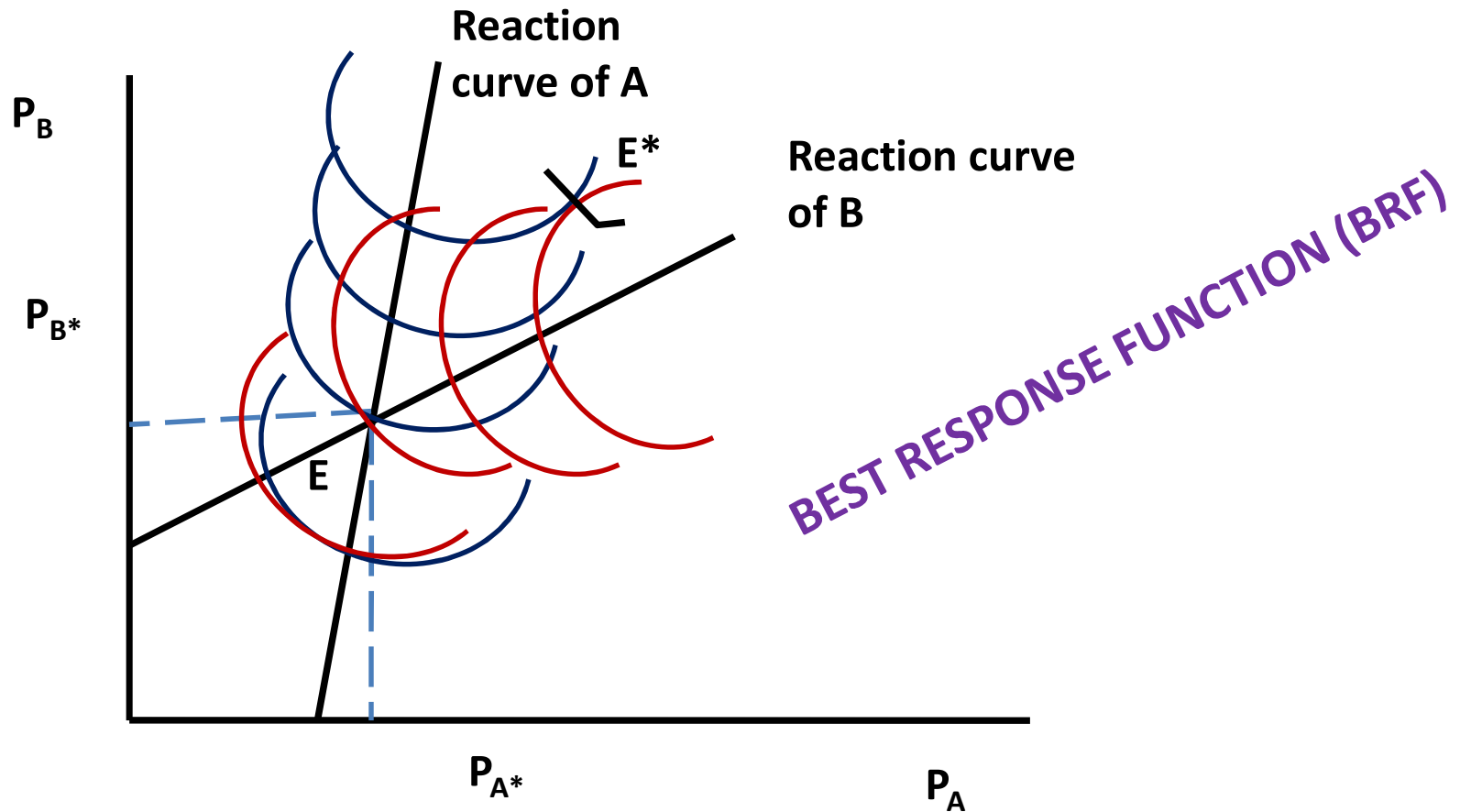
- Convex to the axis along which we measure price of the firm.
- Isoprofit curve far from the axis represent higher profit.
- By connecting each minimum point of successive isoprofit curves, we get reaction curve of a firm.

Continued....



Equilibrium of the model

- The model reaches its equilibrium at the intersection of the two reaction curves.



Continued...

- If one firm charges price higher than the equilibrium price there is always action and reaction that leads to equilibrium.
- ***At Bertrand equilibrium, profit of each firm is maximized and joint profit of the firms is not maximized.***
- Had they made agreement, they can get higher profit from the current output by charging higher price at the tangency point of higher isoprofit curves of the two firms.

Bertrand model with differentiated products

- **Products are imperfect substitutes**
- Demand for each firm is different.
- Firms have market power due to product heterogeneity.
- **Q1 and q2.**
- ***At Equilibrium***
 - ***MR1=MC1***
 - ***MR2=MC2***

Bertrand duopoly with asymmetric cost

- The firm with lower costs can **undercut** the other firm to capture the entire market
 - the lower-cost firm earns positive profits
 - the higher-cost firm earns zero profits.
- **No equilibrium solution if homogenous product.**

Traditional Example: Bertrand

- Two firms selling spring water

- *Inverse demand*

$$-P(Q) = a - Q$$

- Cost

$$-TC_i(q_i) = cq_i$$

– Equilibrium Outcomes

- $Q^* = a - c$

- $p^* = c$

- $\pi^* = 0$

Example: Bertrand model

- **Exampple 1**

- Demand

- $P = 500 - 2Q, Q = q_1 + q_2$

- Cost functions

- $TC_1 = 20q_1$ and $TC_2 = 20q_2$ or $TC=20Q$

- REQUIRED

- Equilibrium price and quantity of each fim and the mrket.

Example: Bertrand model

- **Example 2.**

- Suppose that the demand function is for two imperfectly substitute goods as;

- $q_1 = 400 - P_1 + 2P_2$, $q_2 = 500 - 3P_2 + P_1$ and that the firms total cost functions are

- **Cost**

- *Cost of firm 1 ... $MC_1 = 100$ Cost of firm 2... $MC_2 = 100$*

- **REQUIRED**

- Equilibrium price and quantity of each firm and the market.

- **Example 3.**

- $Q_1 = 1400 + 100P_2 - 200P_1$, and $Q_2 = 1400 + 100P_1 - 200P_2$

- **MC=2**

Answer

- **1. Same MC(identical product)**
 - $P = MC, q1 = q2 \text{ or } Q = 2q1 \text{ or } Q = 2q2$
- **2 Differentiated product**
 - Different demand
 - $P1=MC1$ and $p2=MC2$ OR
 - Derivative of profit w.r.t. p .

Stackelberg Duopoly Model

- Developed by **Heinrich von Stackelberg**, a **German economist (1905 – 1946)**
 - First Economist who systematically studied leader-follower interactions.
 - Published his influential work on market organization in 1934.
- It is similar with Cournot reaction curve approach
- **The Stackelberg model is often used to describe industries in which there is a dominant firm, or a natural leader.**
- **This is essentially a two-stage model in which one firm gets to move first.**
- The other firm can then observe the first firm's output choice and choose its own optimal level of output.
 - **Hence, sequential game**

Stackelberg Oligopoly Model

- It is similar with Cournot reaction curve approach
- All the assumptions, isoprofit curves, and reaction curves are the same except,
 - I) in this model one firm is naïve and follower, and the other is ***sophisticated and leader***.
 - II) The leader knows every thing about the naïve and how the naïve react against its decision.
 - III) So the leader set its equilibrium quantity on the most lower isoprofit curve tangent to the reaction curve of the naïve firm.
 - IV) IV) Equilibrium decision is not simultaneously like Cournot but, sequentially.

The equilibrium

- On the most lower isoprofit that represent higher profit to the leader
- Tangent to the reaction curve of the naïve.
- If the equilibrium point is on the reaction curve of the naïve firm, then since the naïve takes it as equilibrium it does not react against the equilibrium

Equilibrium of the model

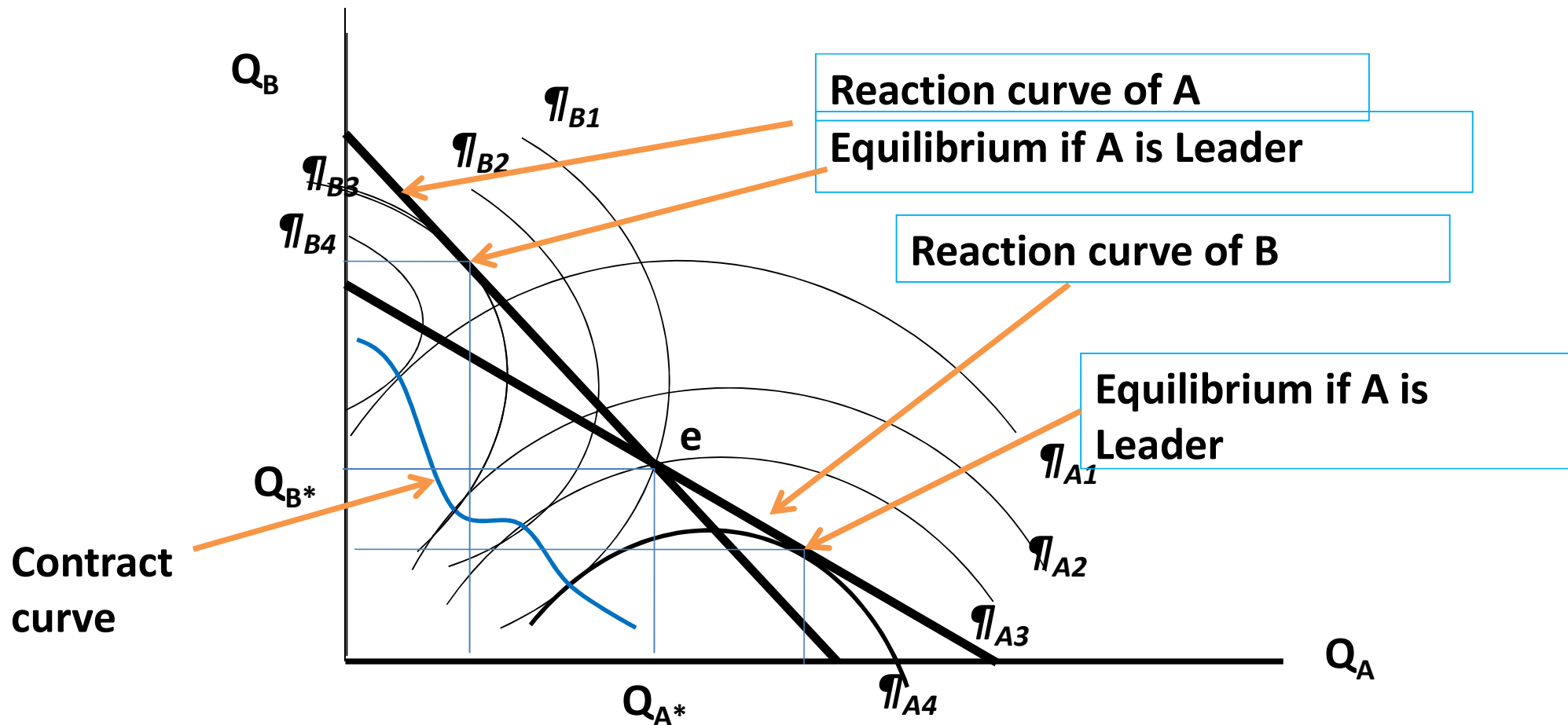


Figure 2.4.: cournot's duopoly model equilibrium

Continued...

- If both firms are sophisticated, both wants to be a leader to get higher profit.
- In this case, market situation becomes unstable.
- **Such situation is known as Stackleberg disequilibrium.**
- The instability continues until one surrender by price war or make agreement to produce on the contract curve.

Mathematically

- The leaders first determine the reaction function of its follower and then incorporate it to its own profit function.
- *Maximize its profit by setting marginal revenue equals to marginal cost*
- **Example**
 - Assume there are two firms producing homogeneous product and demand equation is $P = 60 - 2(y_1 + y_2)$
 - Cost of firm 1 is $C1 = y_1^2$ and cost of firm 2 is $C2 = 4y_2$

Continued....

- a) Find Stackleberg equilibrium $y_1, y_2, Y, P, \pi_1, \pi_2, \Pi$ if 1 is leader
- b) Find Stackleberg equilibrium $y_1, y_2, Y, P, \pi_1, \pi_2, \Pi$ if B is leader

Solution

- a) $y_1=8, y_2=10, Y=18, P=24, \pi_1=128, \pi_2=300, \Pi=428$
- b) $y_1=5.5, y_2=13.5, Y=19, P=22, \pi_1=209, \pi_2=90.75, \Pi=299.75$

Collusive oligopoly

- Collusion is the implicit or explicit (formal or informal) agreement between firms on decision making
- Advantages of collusion
 - it increase profit by decreasing competition
 - reduce uncertainty
 - prevent new entrants that increase market share of the collusion

Cartel

- A cartel is a formal organization of firms producing the same product
- Its objective is to maximize the firms' joint profit
- The followings are examples of cartels
 - a) Cartels aiming at joint profit maximization**
 - b) Market sharing cartel**

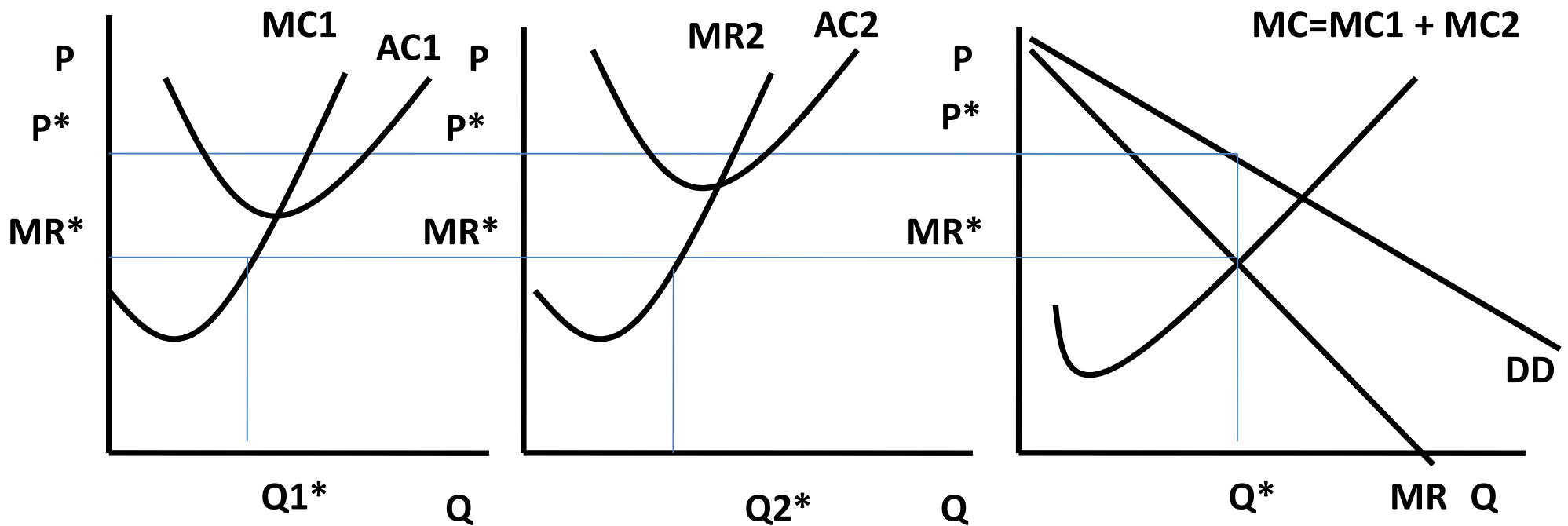
Cartels aiming at joint profit maximization

- Member firms appoint a central agent that works to maximize the joint profit of the cartel
- Cartel knows about the cost of each firm and other information
- Authority of the central agency
 - set equilibrium price , quantity and profit of the cartel
 - allocate the out put among firms
 - allocate the profit among the firms

Continued...

- The central agency carry out its activities in this way
- First estimate the cartel demand
- Derive the cartel marginal revenue (MR)
- Derive cartel marginal cost by summing up MC of each firm i.e. $MC = MC_1 + MC_2$
- Equate the MR and MC and set equilibrium P, Q and calculate profit of the cartel
- Allocate the out put and profit where $MR=MC_1$, and $MR=MC_2$

Continued...



Mathematically

$$\pi = R_1 + R_2 - C_1 - C_2$$

Joint profit function

$$\pi = p(y_1 + y_2) - C_1(y_1) - C_2(y_2)$$

$$\frac{\partial \pi_1}{\partial y_1} = \frac{\partial [P(y_1 + y_2) - c_1(y_1) - c_2(y_2)]}{\partial y_1} = 0 \Rightarrow MR_1 = MC_1$$

$$\frac{\partial \pi_1}{\partial y_1} = \partial \left[\frac{P(y_1 + y_2) - C_2(y_1) - C_2(y_2)}{\partial y_2} \right] = 0 \Rightarrow MR_1 = MC_2$$

$$MR = MR_1 = MR_2 = MC_1 = MC_2$$

Example

Assume demand function of a Cartel having two members is $P=60-2Y$ and cost of firm 1 is $C_1=y_1^2$ and Cost of the second firm $C_2=4y_2$

- a) Find equilibrium y_1, y_2, Y, P ,
- b) Find equilibrium π_1, π_2 , and π

Solution

- a) $y_1=2, y_2=12, Y=14, P=32$
- b) $\pi_1=, \pi_2=$, and $\pi=396$

Continued.....

Reading assignment

- Cartel joint profit is rarely achieved . What are the reasons?

Market Sharing Cartel

- In such cartel firms make agreement on how to share the market

I) **Non- price competition**

- Firms set common price through bargaining and can compete by differentiating their product
- The low cost firm usually violate and it is not stable cartel

Continued....

II) Sharing of the market by agreement on quotas

- Firms set and agree on how much each firm supply
- Quota depends on cost of a firm and other factors
- If all firms have equal cost, they supply equally
- If their cost is different quota depends on their bargaining power
- During bargaining process past levels sales and or the basis for productive capacity of the firms are considered for decision.

Continued....

III) Regional sharing cartels

- Firms share the market by defining markets (geographical boundary) where each firm supply
- Such cartel is not stable because of
 - The firm with low cost encroach to the area of the other
 - firms reach other firms' market through advertisement

❖ **Reading assignment: Read other factors why cartel is unstable collusion**

Price leadership

- Cartels agreement to cooperate is explicit
- Price leadership agreement to cooperate is implicit with out any formal discussion
- Firms enter in to such agreement voluntarily to avoid any uncertainty about the competitor reaction
- One firm implicitly recognized as the leader and set price and others follow

A) Low cost price leadership

Assumptions

- Two firms
- Homogeneous product
- Different costs
- Firms may have equal or unequal market share
- ❖ The low cost firm become leader and set price
- ❖ The other follow taking the price
- ❖ If the follower refused to take the price the low cost firm eliminate through price competition

Continued...

- But the follower firm can influence the price set by the leader through supplying more or less than the expected amount
- So the leader has to make market share agreement formally or informally
- Thus, the leader set equilibrium price taking the market demand curve as its own and make agreement with the follower how much to produce

Mathematically

- Assume there are two firms producing identical product.
- Market demand: $P = 60 - 2(y_1 + y_2)$
- **Cost** $C(y_1) = y_1^2$ $C(y_2) = 4y_2$
- Assuming firm 2 is low cost firm and they share the market equally,
 - a) Find the equilibrium quantities(y_1 and y_2) and price set by the low cost firm
 - b) Find the profit of the two firms
 - c) Price set by firm 2 does not maximize profit of 1, find price that maximize profit of firm 1

Continued ...

Solution

- Hint: If they share the market equally $y_1=y_2$ and demand equation become $P=60-2(y_1+y_2)$
- a) $y_1=7, P=32,$
- b) $\pi_1=175, \pi_2=196$
- c) $Y=6, p=36$

B) Dominant firm price leadership

Assumptions

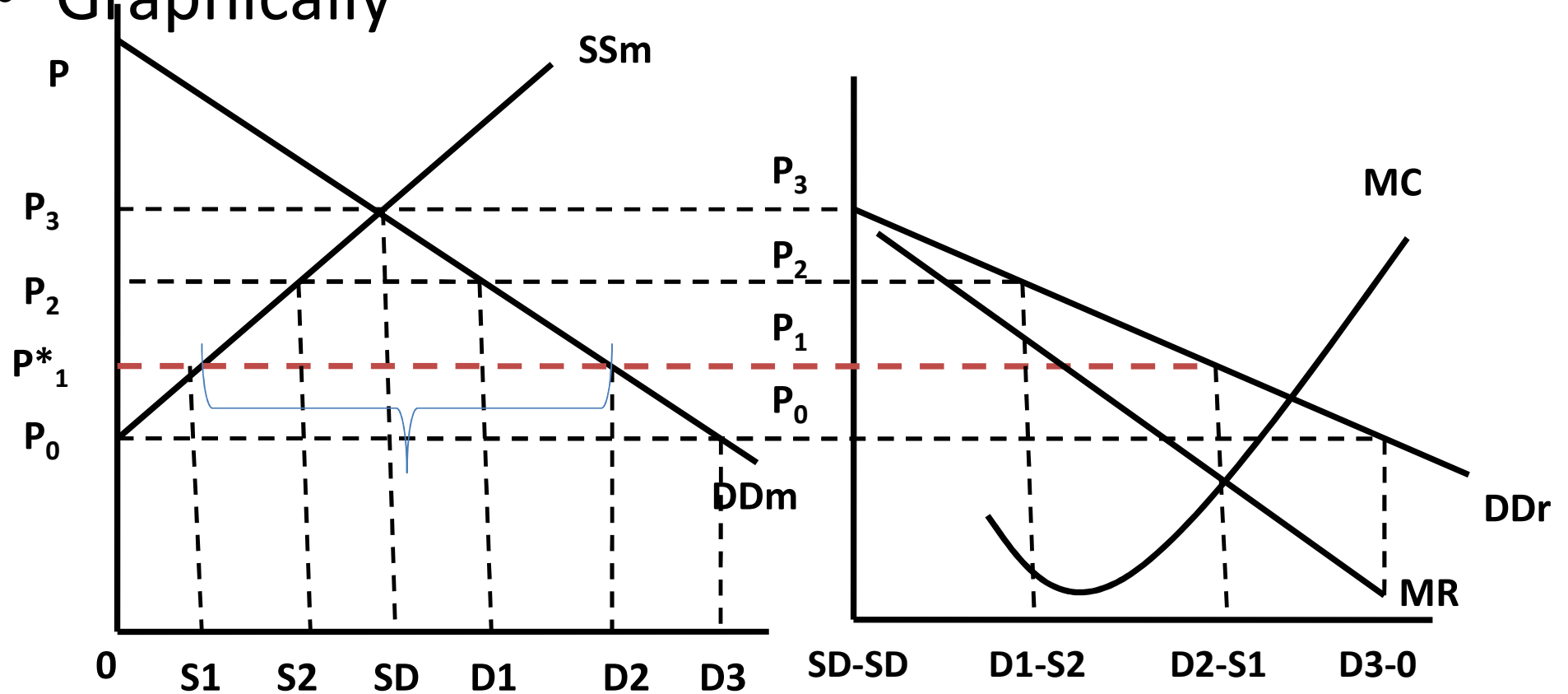
- There are few firms among which one is dominant
- They supply homogeneous product
- The dominant firm knows market demand and MC of small firms
- ❖ The large (dominant firm), that supply more than the others set equilibrium price
- ❖ Small firms behave like perfect competitive firm and follow it
- ❖ How does the dominant firm set equilibrium price?

Continued....

- The dominant firm estimate market demand function
- Derives small firms SS curve by summing up their MC
- Equate market demand and small firms supply curve
- Since small firms act as perfectly competitive firm the leader compare P and summation of MC ($P = \sum MC$)
- By decreasing price, it derives its demand curve (residual demand curve)
- Residual DD = DD – small firms SS
- Derive its MR
- Set equilibrium P by equating its MR and MC

Continued...

- Graphically



Continued...

- When price is P_3 market demand and market supply are equal, so at this price the dominant firm does not supply.
- At $P=P_2$, $D_1>S_2$, thus, D_1-S_2 is demand for the dominant firm
- At $P=P_1$, $D_2>S_1$, thus, D_2-S_1 is demand for the dominant firm
- At $P=P_0$, $D_3>0$, small firms eliminated from the market and all market buyers demand product of the dominant
- Equilibrium price is set at the point where MR and MC of the dominant intersect (equilibrium Price P^*1)

Mathematically

- Assume there are few firms supplying homogeneous product where there is one dominant firm.
- If the estimated demand function of the market is $D(P)=50-0.3p$, Small firms supply equation $S(P)=0.2P$, and the dominant firm cost function is $C=2Y$ then
 - a) Find the dominant firm residual demand curve equation, MR and MC
 - b) Set equilibrium quantity of dominant and equilibrium P
 - c) what is the total quantity demanded at equilibrium
 - d) Small firms supply at equilibrium
 - e) What is the profit of the dominant?

Continued...

Solution

a) $Y = D(P) - S(P) \rightarrow P = 100 - 2Y$

$$MR = 100 - 4Y$$

$$MC = 2$$

b) $Y = 24.5, P = 51$

c) $D(P) = 34.7$

d) $S(p) = 10.2$

e) $\pi = 1200.5$

C) Barometric price leadership

- a firm which have good knowledge of the prevailing condition in the market and can forecast better than other about the future development in the market.

=====//=====